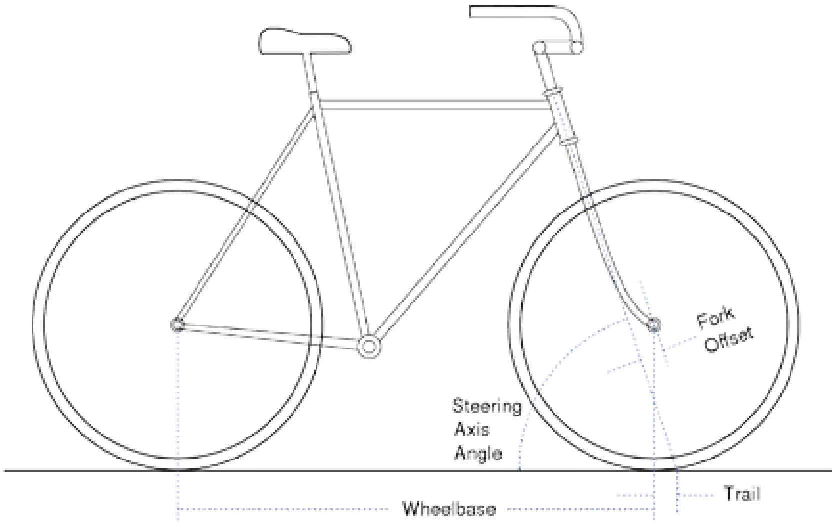
Vesuvius victims
Incredibly preserved remains of a master and his slave have been discovered in Pompeii. **https://dgit.in/dec20-25**

Bezos Earth fund
Jeff Bezos is donating the first chunk of his $10 billion climate change fund to several advocacy groups. **https://dgit.in/dec20-26**

wheel is properly attached to the bike before riding it again.

The second theory is called the caster theory. If you notice the wheels on shopping carts, airport trolleys, or office chairs, when they are moving in a certain direction, all the wheels automatically align themselves in the direction of the motion. This is because the point at where the steering axis meets the ground is a little ahead of the point where the wheels make contact with the ground, creating what is known as the trail. In bikes as well, the forward curve of the fork means that the steering forces are applied slightly ahead of the point at which the wheels make contact to the ground, creating a trail. According to the caster theory, this trail causes the wheels to straighten up, align themselves with the bike, and prevent it from falling even when it is riderless.



The parameters of bike geometry

The research on bicycle dynamics goes back more than a hundred years, and can be said to start with Francis Whipple from Cambridge University, who derived the linearised equations to predict the behavior of a bicycle. For this purpose, the bicycle was abstracted into mathematics, with a model that reduces the bike to four objects. These objects are the front and back wheels, the frame of the bike (which is considered to be rigid), and the steering assembly. The equations derived by Whipple are similar to the equations used in modern times to probe bicycle dynamics. The gyro and caster effect provide the theoretical background for explaining some of the behaviors of bicycles, but how to test if the theories are right?

PUTTING THE THEORIES TO THE TEST

The simplest answer is to build bikes specifically to test the validity of the theories. Dr Hugh Hunt, also from Cambridge University built a bicycle that had a second front wheel attached to the main front wheel. This front wheel had the rim, but no tyre. It could be turned forwards or backwards at different speeds independently of the main front wheel. Additionally, it was mounted on one side of the front wheel, theoretically compromising on the balance. However, no matter in what direction the second front wheel was turned, and how fast it was turned, the bike remained stable and unaffected by the gyroscopic forces of the second spinning wheel.

Andy Ruina from Cornell University along with Arend Schwab from Tu Delft built a bike without a trail, and really tiny wheels to get rid of both the gyroscopic and caster effect. This bike was a physical version of what can be considered as a bike mathematically, a weird hybrid between a foot scooter and a bicycle frame. This "bike" also manages to be stable in a riderless condition which indicates that neither the gyro nor the caster theories are right.

Credit: Hugh Hunt

The gyro bike with the additional front wheel

Richard E Klein from the University of Illinois has built a number of bikes where either the caster effect or the gyro effect has been removed entirely from the bike. This is accomplished by including two sets of wheels on both the front and the back. Along with the regular wheels, there are counterspinning wheels that spin in the opposite direction, removing any of the gyroscopic stabilisation offered by the angular momentum of the wheels. The other approach is have the fork and steering mechanism perpendicular to the ground, so that there is absolutely no trail on the bike. All these experiments have produced odd, but perfectly functional bikes. There are two "naive" bikes, and two Zero-Gyroscope bikes. All these experiments goes to show that neither the gyro nor the caster theories are correct.

So what really causes the bicycle to correct its own course, and be stable even when both the gyro and coaster effects are removed from the bike? This is a matter of active research, and scientists are still trying to figure it out! Maybe CS Lewis was right after all, and there is an invisible wall that prevents bikes from falling over.